

Regression Analysis

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1. Introduction

1.1. Project Libraries and Installation

```
# Project libraries
libs <- c("genridge", "tidyverse", "skimr", "corrplot", "car",
          "lmtest", "glmnet", "MASS", "performance", "see", "gt", "psych")
```

```
# Install only the missing ones
missing <- setdiff(libs, rownames(installed.packages()))
if (length(missing) > 0) {
  install.packages(missing, repos = "https://cloud.r-project.org")
}

# Load all of them
invisible(lapply(libs, library, character.only = TRUE))
```

1.2. Dataset Analysis

1.2.1. Importing the dataset

The dataset to be used in this activity will be `prostate` from the `genridge` library. This dataset contains the following structure:

```
# Information about the library
library(genridge)
?prostate
```

1.2.2. Dataset

```
# Dataframe creation
data(prostate, package="genridge")
df <- prostate |> dplyr::select(-train)
# Stringify dataset
str(df)
```

```
'data.frame':  97 obs. of  9 variables:
 $ lcvol : num  -0.58 -0.994 -0.511 -1.204 0.751 ...
 $ lweight: num  2.77 3.32 2.69 3.28 3.43 ...
 $ age   : int   50 58 74 58 62 50 64 58 47 63 ...
 $ lbph  : num  -1.39 -1.39 -1.39 -1.39 -1.39 ...
 $ svi   : int    0 0 0 0 0 0 0 0 0 0 ...
 $ lcp   : num  -1.39 -1.39 -1.39 -1.39 -1.39 ...
 $ gleason: int   6 6 7 6 6 6 6 6 6 6 ...
 $ pgg45 : int   0 0 20 0 0 0 0 0 0 0 ...
 $ lpsa  : num  -0.431 -0.163 -0.163 -0.163 0.372 ...
```

```
# Dataset summary
print(summary(df))
```

lcavol	lweight	age	lbph
Min. : -1.3471	Min. : 2.375	Min. : 41.00	Min. : -1.3863
1st Qu.: 0.5128	1st Qu.: 3.376	1st Qu.: 60.00	1st Qu.: -1.3863
Median : 1.4469	Median : 3.623	Median : 65.00	Median : 0.3001
Mean : 1.3500	Mean : 3.629	Mean : 63.87	Mean : 0.1004
3rd Qu.: 2.1270	3rd Qu.: 3.876	3rd Qu.: 68.00	3rd Qu.: 1.5581
Max. : 3.8210	Max. : 4.780	Max. : 79.00	Max. : 2.3263

svi	lcp	gleason	pgg45
Min. : 0.0000	Min. : -1.3863	Min. : 6.000	Min. : 0.00
1st Qu.: 0.0000	1st Qu.: -1.3863	1st Qu.: 6.000	1st Qu.: 0.00
Median : 0.0000	Median : -0.7985	Median : 7.000	Median : 15.00
Mean : 0.2165	Mean : -0.1794	Mean : 6.753	Mean : 24.38
3rd Qu.: 0.0000	3rd Qu.: 1.1787	3rd Qu.: 7.000	3rd Qu.: 40.00
Max. : 1.0000	Max. : 2.9042	Max. : 9.000	Max. : 100.00

lpsa
Min. : -0.4308
1st Qu.: 1.7317
Median : 2.5915
Mean : 2.4784
3rd Qu.: 3.0564
Max. : 5.5829

```
# Skim through the data
skimr::skim(df)
```

Name	df
Number of rows	97
Number of columns	9

Column type frequency:

numeric	9
---------	---

Group variables

None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
lcavol	0	1	1.35	1.18	-1.35	0.51	1.45	2.13	3.82	
lweight	0	1	3.63	0.43	2.37	3.38	3.62	3.88	4.78	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	63.87	7.45	41.00	60.00	65.00	68.00	79.00	
lbph	0	1	0.10	1.45	-1.39	-1.39	0.30	1.56	2.33	
svi	0	1	0.22	0.41	0.00	0.00	0.00	0.00	1.00	
lcp	0	1	-0.18	1.40	-1.39	-1.39	-0.80	1.18	2.90	
gleason	0	1	6.75	0.72	6.00	6.00	7.00	7.00	9.00	
pgg45	0	1	24.38	28.20	0.00	0.00	15.00	40.00	100.00	
lpsa	0	1	2.48	1.15	-0.43	1.73	2.59	3.06	5.58	

```
# Head of the dataset
print(head(df))
```

```
      lcavol  lweight age      lbph svi      lcp gleason pgg45      lpsa
1 -0.5798185 2.769459 50 -1.386294  0 -1.386294      6      0 -0.4307829
2 -0.9942523 3.319626 58 -1.386294  0 -1.386294      6      0 -0.1625189
3 -0.5108256 2.691243 74 -1.386294  0 -1.386294      7     20 -0.1625189
4 -1.2039728 3.282789 58 -1.386294  0 -1.386294      6      0 -0.1625189
5  0.7514161 3.432373 62 -1.386294  0 -1.386294      6      0  0.3715636
6 -1.0498221 3.228826 50 -1.386294  0 -1.386294      6      0  0.7654678
```

```
# Tail of the dataset
print(tail(df))
```

```
      lcavol  lweight age      lbph svi      lcp gleason pgg45      lpsa
92 2.532903 3.677566 61  1.3480732  1 -1.386294      7     15 4.129551
93 2.830268 3.876396 68 -1.3862944  1  1.321756      7     60 4.385147
94 3.821004 3.896909 44 -1.3862944  1  2.169054      7     40 4.684443
95 2.907447 3.396185 52 -1.3862944  1  2.463853      7     10 5.143124
96 2.882564 3.773910 68  1.5581446  1  1.558145      7     80 5.477509
97 3.471966 3.974998 68  0.4382549  1  2.904165      7     20 5.582932
```

```
# Shape / Dimensions of the Dataset
print(dim(df))
```

```
[1] 97  9
```

1.3. Descriptive Statistics

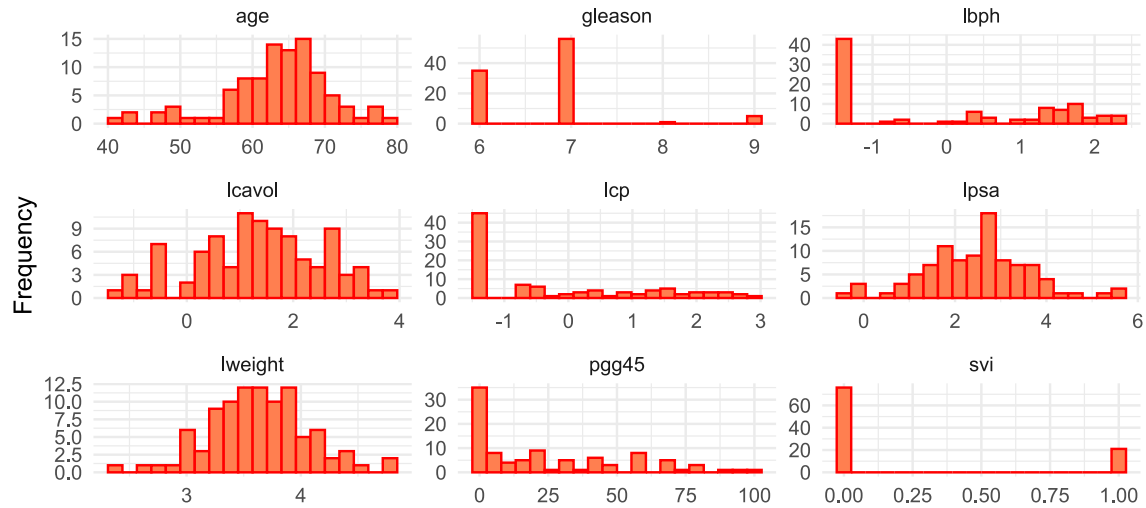
```
# Descriptive statistics with psych
psych::describe(df) |>
  as.data.frame() |>
  dplyr::select(n, mean, sd, median, min, max, range, skew, kurtosis, se) |>
  round(2) |>
  gt::gt(rownames_to_stub = TRUE)
```

	n	mean	sd	me- dian	min	max	range	skew	kur- tosis	se
lcavol	97	1.35	1.18	1.45	-1.35	3.82	5.17	-0.24	-0.60	0.12
lweight	97	3.63	0.43	3.62	2.37	4.78	2.41	0.06	0.36	0.04
age	97	63.87	7.45	65.00	41.00	79.00	38.00	-0.80	0.96	0.76
lbph	97	0.10	1.45	0.30	-1.39	2.33	3.71	0.13	-1.75	0.15
svi	97	0.22	0.41	0.00	0.00	1.00	1.00	1.36	-0.16	0.04
lcp	97	-0.18	1.40	-0.80	-1.39	2.90	4.29	0.71	-1.01	0.14
glea- son	97	6.75	0.72	7.00	6.00	9.00	3.00	1.22	2.36	0.07
pgg45	97	24.38	28.20	15.00	0.00	100.00	100.00	0.94	-0.37	2.86
lpsa	97	2.48	1.15	2.59	-0.43	5.58	6.01	0.00	0.43	0.12

1.3.1. Histograms

```
df |>
  pivot_longer(everything(), names_to = "variable", values_to = "value") |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 20, fill = "coral", color = "red") +
  facet_wrap(~ variable, scales = "free") +
  labs(title = "Distribution of every variable", x = NULL, y = "Frequency") +
  theme_minimal()
```

Distribution of every variable



1.3.2 Normality Test

Shapiro-Wilk test for normality on the variables in the dataset.

```
# Shapiro-Wilk on lcavol
shapiro.test(df$lcavol)
```

Shapiro-Wilk normality test

```
data: df$lcavol
W = 0.98295, p-value = 0.2414
```

```
# Shapiro-Wilk on lweight
shapiro.test(df$lweight)
```

Shapiro-Wilk normality test

```
data: df$lweight
W = 0.9936, p-value = 0.929
```

```
# Shapiro-Wilk on age
shapiro.test(df$age)
```

Shapiro-Wilk normality test

```
data: df$age  
W = 0.94679, p-value = 0.0006376
```

```
# Shapiro-Wilk on lbph  
shapiro.test(df$lbph)
```

Shapiro-Wilk normality test

```
data: df$lbph  
W = 0.79376, p-value = 2.454e-10
```

```
# Shapiro-Wilk on svi  
shapiro.test(df$svi)
```

Shapiro-Wilk normality test

```
data: df$svi  
W = 0.50717, p-value < 2.2e-16
```

```
# Shapiro-Wilk on lcp  
shapiro.test(df$lcp)
```

Shapiro-Wilk normality test

```
data: df$lcp  
W = 0.80618, p-value = 5.842e-10
```

```
# Shapiro-Wilk on gleason  
shapiro.test(df$gleason)
```

Shapiro-Wilk normality test

```
data: df$gleason  
W = 0.70326, p-value = 1.049e-12
```

```
# Shapiro-Wilk on pgg45  
shapiro.test(df$pgg45)
```

Shapiro-Wilk normality test

```
data: df$pgg45  
W = 0.82336, p-value = 2.065e-09
```

```
# Shapiro-Wilk on lpsa  
shapiro.test(df$lpsa)
```

Shapiro-Wilk normality test

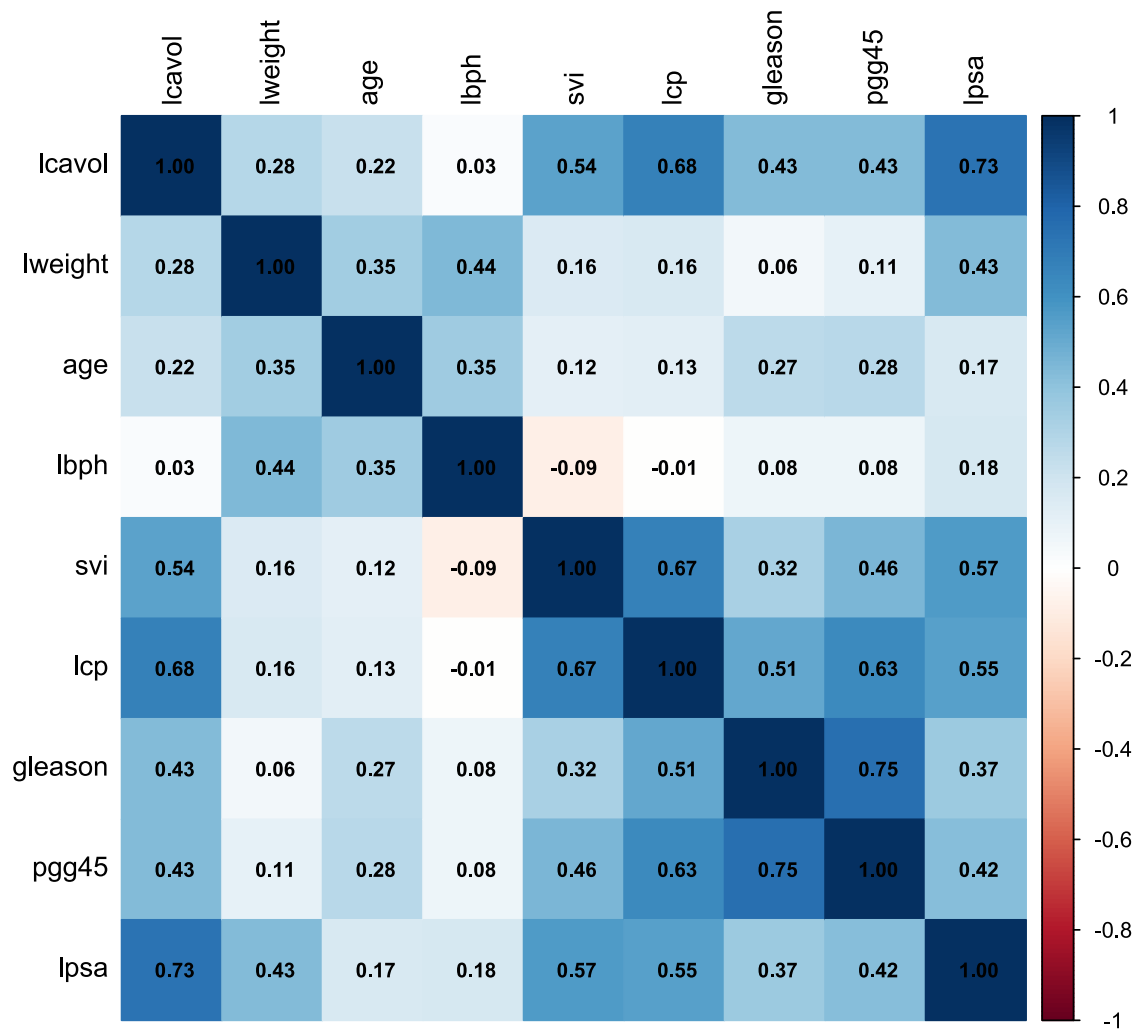
```
data: df$lpsa  
W = 0.98443, p-value = 0.3087
```

Observations:

- The variables that follow a **normal distribution** according to the results given by the Shapiro-Wilk test are: `lcavol`, `lweight`, and `lpsa`. The rest of variables **do not** follow a normal distribution.

1.4. Correlation Matrix

```
corrplot(cor(df), method = "color", addCoef.col = "black",  
         number.cex = 0.75, tl.col = "black")
```



2. Train / Test Split

The seed is fixed with `set.seed(123)` so the split is reproducible. 70% of the observations are used to train the models and the remaining 30% to evaluate their performance.

```
set.seed(123)
idx <- sample(nrow(df), 0.7 * nrow(df))
train <- df[idx, ]
test <- df[-idx, ]

cat("Train:", nrow(train), "observations\n")
```

```
Train: 67 observations
```

```
cat("Test: ", nrow(test), "observations\n")
```

```
Test: 30 observations
```

3. Multiple Linear Regression (Full Model)

lpsa is the response variable and all the other variables are predictors.

```
full_model <- lm(lpsa ~ ., data = train)
summary(full_model)
```

Call:

```
lm(formula = lpsa ~ ., data = train)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.70137	-0.32901	-0.03892	0.43860	1.60677

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1.193679	1.958227	-0.610	0.54453	
lcavol	0.466150	0.105449	4.421	4.38e-05	***
lweight	0.723549	0.261043	2.772	0.00748	**
age	-0.013025	0.014049	-0.927	0.35771	
lbph	0.073486	0.067706	1.085	0.28225	
svi	0.669491	0.292442	2.289	0.02572	*
lcp	-0.088718	0.111805	-0.794	0.43071	
gleason	0.145557	0.216649	0.672	0.50434	
pgg45	0.004901	0.005492	0.892	0.37587	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6956 on 58 degrees of freedom

Multiple R-squared: 0.6333, Adjusted R-squared: 0.5827

F-statistic: 12.52 on 8 and 58 DF, p-value: 3.096e-10

3.1. Multicollinearity (VIF)

```
vif(full_model)
```

	lcavol	lweight	age	lbph	svi	lcp	gleason	pgg45
	2.126434	1.379956	1.307110	1.293350	2.242169	3.231911	2.562448	3.034720

Observations:

- Only `lcavol`, `lweight` and `svi` are significant at the 5% level; the other predictors do not contribute significantly once the rest are in the model.
- The highest VIFs correspond to `lcp` and `pgg45`. No VIF is above 5, so multicollinearity is moderate, but the correlated predictors (`lcavol`–`lcp`, `gleason`–`pgg45`) justify applying variable selection methods.

4. Variable Selection Methods

4.1. Forward, Backward and Stepwise

All three methods use the AIC as the selection criterion. Forward and stepwise start from the null model (intercept only); backward starts from the full model.

```
null_model <- lm(lpsa ~ 1, data = train)

forward_model <- step(null_model, scope = formula(full_model),
                      direction = "forward", trace = 0)
backward_model <- step(full_model, direction = "backward", trace = 0)
stepwise_model <- step(null_model, scope = formula(full_model),
                      direction = "both", trace = 0)

formula(forward_model)
```

```
lpsa ~ lcavol + lweight + svi + gleason
```

```
formula(backward_model)
```

```
lpsa ~ lcavol + lweight + svi + pgg45
```

```
formula(stepwise_model)
```

```
lpsa ~ lcavol + lweight + svi + gleason
```

4.2. Lasso, Ridge and Elastic Net

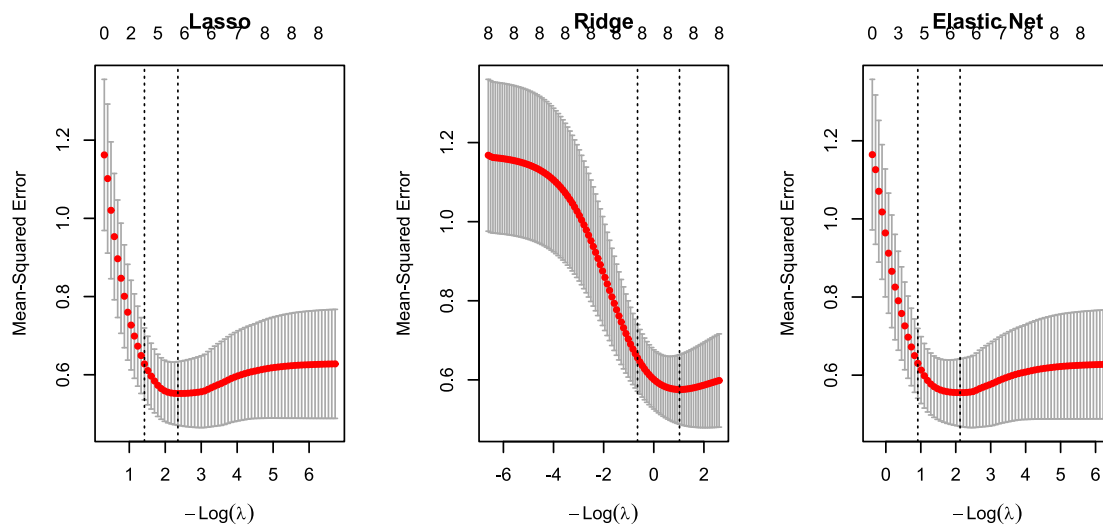
`glmnet` requires a numeric matrix of predictors. The penalty parameter λ is chosen with 10-fold cross-validation (`cv.glmnet`), using the value that minimizes the cross-validation error (`lambda.min`). The seed is set before each call so the folds are reproducible.

- **Lasso:** $\alpha = 1$
- **Ridge:** $\alpha = 0$
- **Elastic Net:** $\alpha = 0.5$

```
x_train <- model.matrix(lpsa ~ ., data = train)[, -1]
y_train <- train$lpsa
x_test  <- model.matrix(lpsa ~ ., data = test)[, -1]

set.seed(123)
lasso_cv <- cv.glmnet(x_train, y_train, alpha = 1)
set.seed(123)
ridge_cv <- cv.glmnet(x_train, y_train, alpha = 0)
set.seed(123)
enet_cv  <- cv.glmnet(x_train, y_train, alpha = 0.5)

par(mfrow = c(1, 3))
plot(lasso_cv, main = "Lasso")
plot(ridge_cv, main = "Ridge")
plot(enet_cv,  main = "Elastic Net")
```



```
par(mfrow = c(1, 1))
```

Coefficients of each penalized model (a coefficient of 0 means the variable was eliminated):

```
penalized_coefs <- cbind(
  as.matrix(coef(lasso_cv, s = "lambda.min")),
  as.matrix(coef(ridge_cv, s = "lambda.min")),
  as.matrix(coef(enet_cv, s = "lambda.min"))
)
colnames(penalized_coefs) <- c("Lasso", "Ridge", "Elastic Net")
round(penalized_coefs, 4)
```

	Lasso	Ridge	Elastic Net
(Intercept)	-0.8405	-1.3571	-1.2444
lcavol	0.3815	0.3034	0.3720
lweight	0.5490	0.5913	0.5941
age	0.0000	-0.0009	0.0000
lbph	0.0000	0.0439	0.0156
svi	0.4705	0.4754	0.5007
lcp	0.0000	0.0462	0.0000
gleason	0.1023	0.1781	0.1373
pgg45	0.0017	0.0028	0.0022

5. Model Comparison (RMSE and MAE)

All models are evaluated on the **test set**, which was not used during training.

```
rmse <- function(y, pred) sqrt(mean((y - pred)^2))
mae <- function(y, pred) mean(abs(y - pred))

predictions <- list(
  "Full model" = predict(full_model, test),
  "Forward"    = predict(forward_model, test),
  "Backward"   = predict(backward_model, test),
  "Stepwise"   = predict(stepwise_model, test),
  "Lasso"      = as.numeric(predict(lasso_cv, x_test, s = "lambda.min")),
  "Ridge"      = as.numeric(predict(ridge_cv, x_test, s = "lambda.min")),
  "Elastic Net" = as.numeric(predict(enet_cv, x_test, s = "lambda.min"))
)

metrics <- tibble(
  Model = names(predictions),
  RMSE  = map_dbl(predictions, function(p) rmse(test$lpsa, p)),
  MAE   = map_dbl(predictions, function(p) mae(test$lpsa, p))
)
```

```
metrics |>
  arrange(RMSE) |>
  gt() |>
  fmt_number(columns = c(RMSE, MAE), decimals = 4)
```

Model	RMSE	MAE
Full model	0.7422	0.5332
Backward	0.7665	0.5776
Elastic Net	0.7980	0.5857
Lasso	0.7994	0.5755
Forward	0.8011	0.6153
Stepwise	0.8011	0.6153
Ridge	0.8261	0.6136

Observations:

- Among the variable selection methods, **Backward** has the lowest RMSE, and its MAE is practically tied with Lasso's (difference smaller than 0.003). It is therefore the method with the best overall performance, and it is the model analyzed in the next section.
- The full model obtains slightly lower errors than all the selection methods. With only 30 test observations these differences are small; Backward achieves similar performance with half of the predictors (`lcavol`, `lweight`, `svi`, `pgg45`), which makes it simpler to interpret.

6. Analysis of the Best Model (Backward)

```
best_model <- backward_model
summary(best_model)
```

Call:

```
lm(formula = lpsa ~ lcavol + lweight + svi + pgg45, data = train)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.65921	-0.41224	-0.07317	0.44640	1.53256

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
----------	------------	---------	----------

```

(Intercept) -1.003776    0.810811   -1.238    0.22039
lcavol      0.426077    0.086718    4.913 6.87e-06 ***
lweight     0.736437    0.226935    3.245 0.00189 **
svi         0.531448    0.250011    2.126 0.03752 *
pgg45       0.005802    0.003595    1.614 0.11161
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6893 on 62 degrees of freedom
Multiple R-squared:  0.6152,    Adjusted R-squared:  0.5903
F-statistic: 24.78 on 4 and 62 DF,  p-value: 2.795e-12

```

6.1. Significance of the Model (ANOVA and F Statistic)

- H_0 : all the coefficients of the predictors are equal to zero.
- H_1 : at least one coefficient is different from zero.

```
anova(best_model)
```

Analysis of Variance Table

Response: lpsa

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
lcavol	1	36.397	36.397	76.6139	1.975e-12	***
lweight	1	5.275	5.275	11.1031	0.001457	**
svi	1	4.173	4.173	8.7831	0.004309	**
pgg45	1	1.238	1.238	2.6049	0.111611	
Residuals	62	29.455	0.475			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

f <- summary(best_model)$fstatistic
p_value <- pf(f["value"], f["numdf"], f["dendf"], lower.tail = FALSE)
cat("F =", round(f["value"], 3), "on", f["numdf"], "and", f["dendf"],
    "DF, p-value =", signif(p_value, 3), "\n")

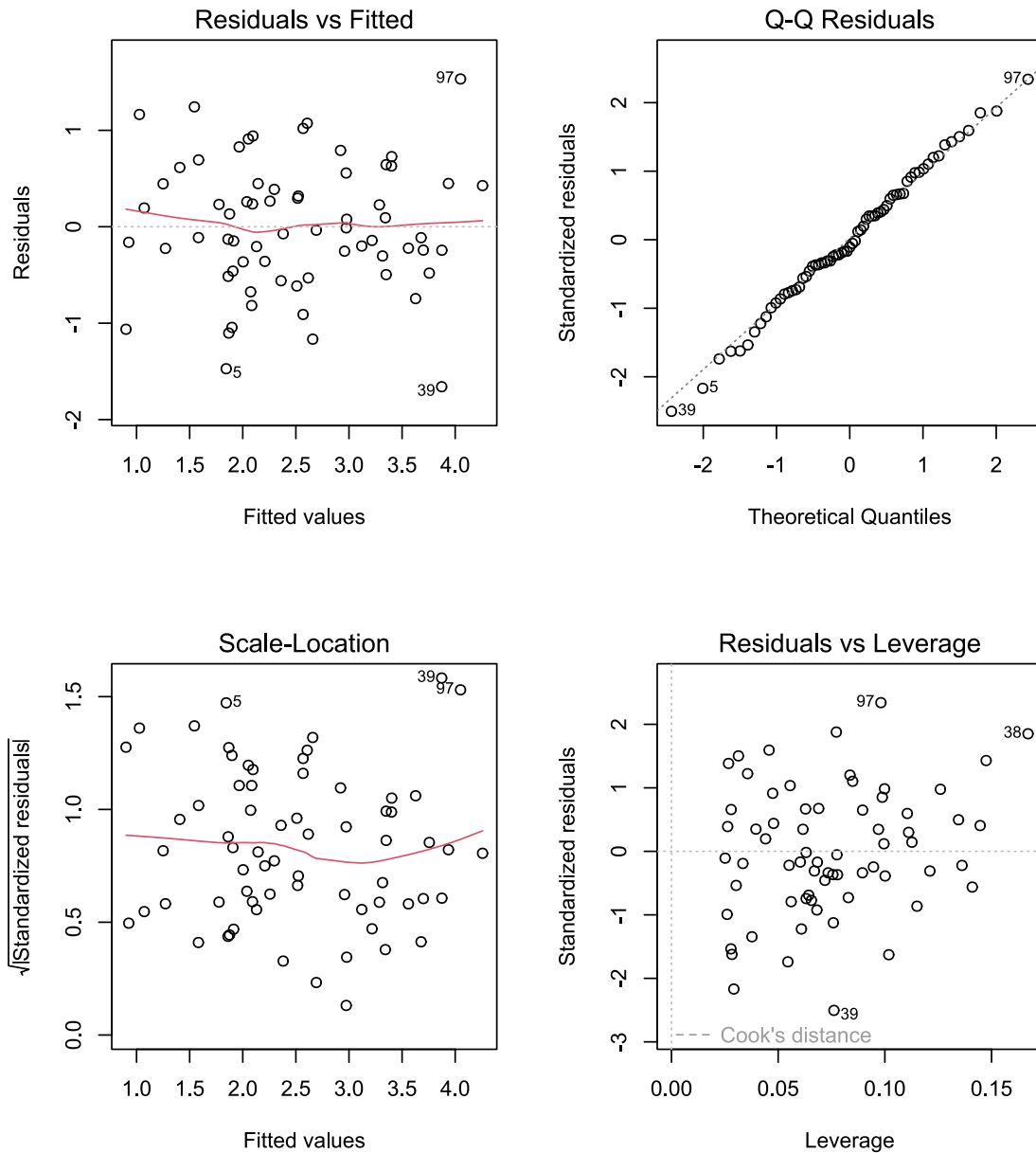
```

F = 24.776 on 4 and 62 DF, p-value = 2.79e-12

Observations: the p-value of the F statistic is far below 0.05, so H_0 is rejected: the model is globally significant. In the ANOVA table, `lcavol`, `lweight` and `svi` are significant, while `pgg45` is not significant at the 5% level, although the AIC kept it in the model.

6.2. Assumption Checks

```
par(mfrow = c(2, 2))  
plot(best_model)
```

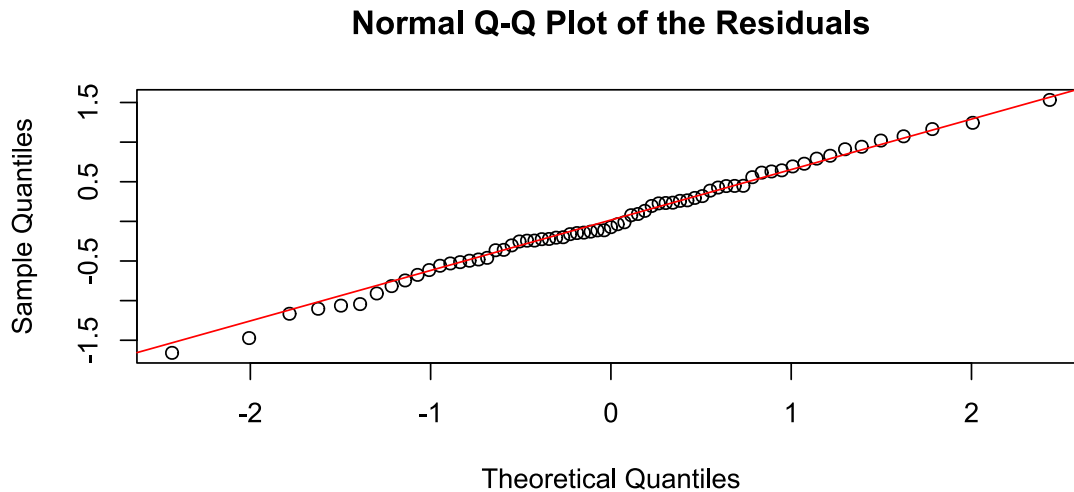


```
par(mfrow = c(1, 1))
```

6.2.1. Normality of the Residuals

- H_0 : the residuals follow a normal distribution.
- H_1 : the residuals do not follow a normal distribution.

```
qqnorm(residuals(best_model), main = "Normal Q-Q Plot of the Residuals")  
qqline(residuals(best_model), col = "red")
```



```
shapiro.test(residuals(best_model))
```

Shapiro-Wilk normality test

```
data: residuals(best_model)  
W = 0.99391, p-value = 0.9862
```

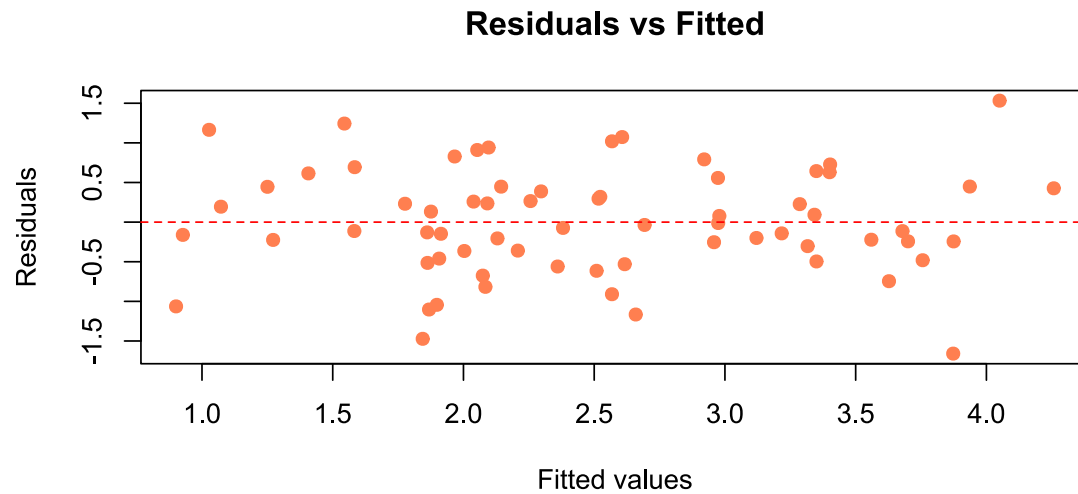
Observations: the points of the Q-Q plot follow the reference line and the p-value of the Shapiro-Wilk test is greater than 0.05, so H_0 is not rejected: there is no evidence against the normality of the residuals.

6.2.2. Homoscedasticity

- H_0 : the variance of the residuals is constant (homoscedasticity).
- H_1 : the variance of the residuals is not constant (heteroscedasticity).

```
plot(fitted(best_model), residuals(best_model),  
     xlab = "Fitted values", ylab = "Residuals",
```

```
main = "Residuals vs Fitted", pch = 19, col = "coral")
abline(h = 0, col = "red", lty = 2)
```



```
bptest(best_model)
```

studentized Breusch-Pagan test

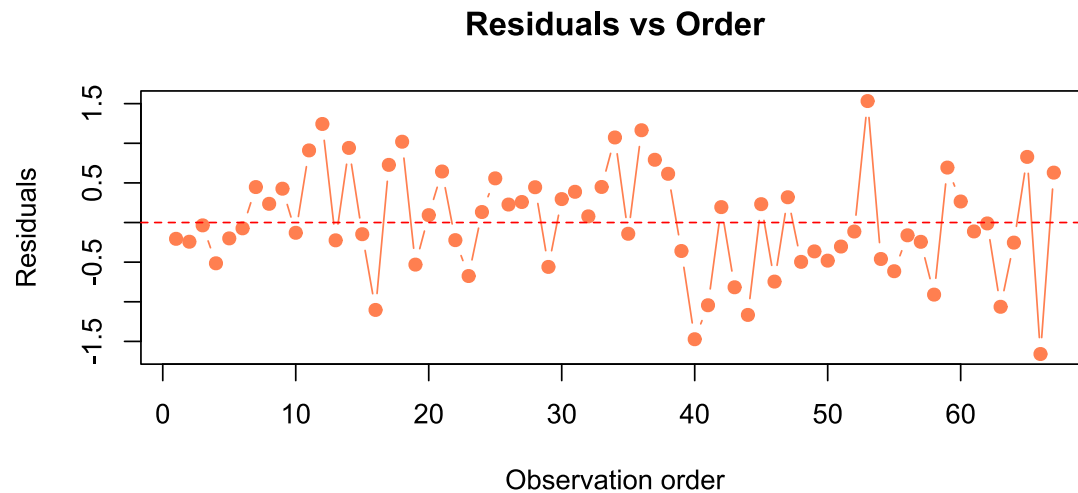
```
data: best_model
BP = 5.5328, df = 4, p-value = 0.2369
```

Observations: the residuals are randomly scattered around zero without a funnel shape, and the p-value of the Breusch-Pagan test is greater than 0.05, so H_0 is not rejected: the variance of the residuals is constant.

6.2.3. Independence

- H_0 : the residuals are not autocorrelated (independence).
- H_1 : the residuals are autocorrelated.

```
plot(residuals(best_model), type = "b",
     xlab = "Observation order", ylab = "Residuals",
     main = "Residuals vs Order", pch = 19, col = "coral")
abline(h = 0, col = "red", lty = 2)
```



```
dwtest(best_model)
```

Durbin-Watson test

```
data: best_model
DW = 1.9191, p-value = 0.3618
alternative hypothesis: true autocorrelation is greater than 0
```

Observations: the residuals show no pattern with respect to the observation order, and the Durbin-Watson statistic is close to 2 with a p-value greater than 0.05, so H_0 is not rejected: the residuals are independent.

7. Conclusions

- The Backward model, with `lcavol`, `lweight`, `svi` and `pgg45`, had the best overall performance among the variable selection methods on the test set.
- The model is globally significant according to the ANOVA F test.
- The assumptions of normality, homoscedasticity and independence of the residuals are satisfied, so the inference of the model is valid.
- The main predictor of `lpsa` is `lcavol` (the log of the cancer volume), followed by `lweight` and `svi`.